

Tanmay Kala

☎ +91-9530173355 • ✉ tanmaykala171206@gmail.com • 🔗 linkedin.com/in/tanmay-kala • 🌐 github.com/tanmayai23

SUMMARY

Generative AI and NLP engineer with **5 shipped AI products**, **Top-25-of-400+ national hackathon ranking**, and Oracle Cloud GenAI Professional certification. Founder of Naxatra AI. Expert in Python, LLMs, RAG, Prompt Engineering, Hugging Face Transformers, Gemini API, and full-stack AI deployment.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal

Sep 2024 – Sep 2028

B.Tech, Computer Science – Specialization: Artificial Intelligence & Machine Learning

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Database Management Systems, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL, HTML/CSS

AI & Machine Learning: Machine Learning, Deep Learning, Generative AI, Natural Language Processing (NLP), Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Prompt Engineering, Text Classification, Question Answering, Tokenization

Frameworks & Libraries: Hugging Face Transformers, OpenAI API, Gemini API, TensorFlow, PyTorch (basic), scikit-learn, pandas, NumPy, OpenCV

Speech & Vision: Whisper AI, Speech Recognition, Text-to-Speech (pyttsx3), Hotword Detection (Snowboy, Porcupine), Computer Vision

Web & Backend: Flask, FastAPI, Gradio, Streamlit, REST API Development, pywhatkit, pyngrok, Vercel

Databases: SQLite, MySQL, Vector Databases

Cloud & DevOps: Oracle Cloud Infrastructure (OCI), AWS, Google Cloud Platform (GCP), Git, GitHub, Google Colab, Model Deployment, Data Pipelines

PROJECTS

SwachhVan – AI Smart Sanitation Platform — *TypeScript, AI Demand Forecasting, GPS, Vercel* — [Live](#) Feb 2025

- Architected production AI urban sanitation platform; ranked **Top 25 of 400+ teams** at Hack For Green Bharat national hackathon.
- Engineered AI demand forecasting engine cutting manual dispatch planning by **60%** across **3 city zones**.
- Integrated real-time GPS tracking, smart booking, and cleanliness rating into full-stack TypeScript application deployed on Vercel.
- Validated end-to-end AI dispatch pipeline with **100+ simulated bookings**; zero cloud infrastructure cost using Vercel free tier.

Hey Dude – AI Voice Assistant — *Python, Gemini API, OpenCV, Speech Recognition, SQLite*

Nov 2025

- Delivered production AI voice assistant in **15 days**; unified voice recognition, biometric face auth, and system control in 1 Python pipeline.
- Replaced rule-based engine with Gemini API; accelerated conversational AI accuracy **3×** across **10+ command categories**.
- Implemented OpenCV biometric authentication with SQLite face embeddings; achieved passwordless login in **<2 seconds**.
- Automated WhatsApp for **50+ contacts** via voice using pywhatkit; cut task completion time **80%** with **<500 ms** hotword latency (Snowboy + Porcupine).

AI Travel Assist — *Python, Gemini API, SerpAPI, NLP, Geolocation*

Oct 2025

- Developed NLP geolocation AI surfacing hidden POIs within **10 km radius**; auto-categorized into **3 priority states** in real time.
- Integrated Gemini API + SerpAPI; improved recommendation relevance **40%** over baseline search across **3 POI categories**.
- Led sole AI engineering role in **6-member team**; delivered full project within **4-week** academic deadline.

VibeHub – College Life Management Platform — *Python, Flask, Gradio, NLP Chatbot*

Jun 2025

- Shipped student productivity platform consolidating **6 modules** (courses, assignments, attendance, mood, complaints, community).
- Integrated NLP-powered chatbot handling **100+ query types**; eliminated human-agent dependency for platform navigation.
- Delivered Smart Planner and Pomodoro Timer; placed at **Summer of Codefest '25 Hackathon** among **50+ competing teams**.

Whisper AI Transcription Tool — *Python, OpenAI Whisper, Flask, Gradio, pyngrok* Jan 2025

- Deployed audio transcription app achieving **95%+ accuracy** across **5+ formats** via drag-and-drop Gradio interface.
- Added auto language detection and translation for **50+ languages**; generated **3 output formats** (TXT, SRT, JSON).
- Published Colab-hosted app publicly via pyngrok; achieved **zero-cost** production deployment with no cloud infrastructure.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional – <i>Oracle</i>	Oct 2025 – Oct 2027
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – <i>Oracle</i>	Oct 2025
Oracle Cloud Application Foundation Certified Implementation Specialist – <i>Oracle</i>	2024
AI for Beginners – <i>HP LIFE</i>	Jun 2024
MATLAB Onramp – <i>MathWorks Certified</i>	2024

ACHIEVEMENTS & LEADERSHIP

Founder – Naxatra AI 2025 – Present
Building AI-powered products targeting civic and student problems

- Ranked **Top 25 of 400+ teams** nationally at Hack For Green Bharat hackathon with AI sanitation platform SwachhVan.
- Published **21 build-in-public posts** on LinkedIn averaging **538+ impressions each**; grew to **439 followers**.
- Completed **15-day sprint** delivering Hey Dude AI voice assistant from zero to production-ready.
- Maintained **12 public GitHub repositories** spanning Generative AI, NLP, Computer Vision, and full-stack TypeScript.